

CUSTOMER BEHAVIOUR ANALYSIS

End-to-End Data Analytics Project

Python

MySQL

Power BI

5,000

Customers

30

Unique Items

\$994.18 K

Total Spend

3.62

Avg. Review

TABLE OF CONTENTS

01 Project Overview & Objective

02 Domain Background

03 Business Problem Statement

04 Dataset & Data Dictionary

05 Exploratory Data Analysis (Python)

06 SQL Business Analysis

07 Power BI Dashboard

08 Business Recommendations & Conclusion

PROJECT OVERVIEW

Objective: Analyze customer shopping behavior using transactional data from ~5,000 purchases across multiple product categories to uncover purchasing patterns, preferences, and revenue drivers - enabling data-driven business decisions.



5,000

Customer Records



30

Unique Items



3.62

Avg. Review



\$994.18 K

Total Spend



\$198.84

Avg. Spend

Tools Used:

Python

EDA & Data Cleaning

MySQL

Business SQL Analysis

Power BI

Interactive Dashboard

DOMAIN BACKGROUND

Why Customer Behaviour Data Matters in E-Commerce

What Businesses Track

- Products purchased by customers
- Purchase frequency and shopping habits
- Customer spending patterns
- Responses to discounts and promotions
- Preferred payment methods
- Seasonal buying trends

Why This Data Is Important

- Identify high-value and loyal customers
- Understand purchase patterns and preferences
- Develop effective discount strategies
- Improve marketing campaign efficiency
- Enhance customer lifetime value (CLV)
- Reduce customer churn and attrition

BUSINESS PROBLEM STATEMENT

How can a retail e-commerce company better understand customer purchasing behaviour to increase revenue, improve retention, and optimize marketing strategies?

Low Retention

Who are frequent vs. one-time buyers?

Product Demand Gaps

Which categories and products generate highest revenue?

Seasonal Demand

Does season impact sales volume?

Ineffective Discounts

Do discounts increase purchase value or reduce margins?

Payment Behaviour

Which payment methods are preferred?

No Segmentation

How to identify high-value, discount-sensitive customers?

DATASET SUMMARY & DATA DICTIONARY

Type: Retail / E-commerce

Nature: Structured (Tabular)

Level: Customer Transaction

Rows: 5,051 (5,050 + header)

Columns: 17

Post-Cleaning: 5,000 unique records

Key Columns:

Column	Type	Description
Customer ID	Numeric	Unique customer identifier
Purchase Amount	Numeric	Amount spent (USD)
Category	Categorical	Clothing, Electronics, Accessories...
Review Rating	Numeric	Product rating (1-5)
Subscription Status	Categorical	Yes / No
Discount Applied	Categorical	Whether discount was used
Previous Purchases	Numeric	Count of past purchases
Frequency of Purchases	Categorical	Weekly, Monthly, etc.

EXPLORATORY DATA ANALYSIS (PYTHON)

Data Loading

Imported dataset using `pandas.read_csv()`. Explored structure with `df.head()`, `df.info()`, `df.describe()`, `df.isnull().sum()`

Data Consistency Check

Identified mismatches between Item Purchased and Category. Created a mapping dictionary to correct category values

Missing Values - Rating

Imputed using product-level median (not global avg) to keep ratings realistic per item

Missing Values - Size

Domain logic: Electronics/Accessories = Not Applicable, Footwear = Free Size, Clothing = mode

Missing Values - Purchase Amt

Replaced with mean to maintain product-level pricing consistency

Missing Values - Prev. Purchases

Replaced with 0 (implies no prior history) for accurate segmentation

Duplicate Handling

Identified and removed 50 duplicate Customer IDs — final dataset: 5,000 unique records

Column Standardization

Renamed to snake_case using `str.replace()` + `str.lower()` for SQL and Power BI compatibility

Database Integration

Connected to MySQL via SQLAlchemy + pymysql. Exported clean DataFrame using `df.to_sql()`

SQL BUSINESS ANALYSIS - 12 QUERIES

MySQL • Aggregations • Subqueries • Window Functions (RANK, PARTITION BY) • CTEs

Q1

Category Revenue

GROUP BY, SUM, ORDER BY

Q2

Discount Impact

GROUP BY, AVG, SUM

Q3

Gender Revenue Split

GROUP BY gender + category

Q4

High-Spend Discount Users

Subquery + AVG filter

Q5

Top/Bottom 5 by Rating

GROUP BY, AVG, LIMIT

Q6

Shipping Type Comparison

GROUP BY shipping_type

Q7

Subscriber vs Non-Subscriber

GROUP BY subscription_status

Q8

Discount Dependency %

CASE WHEN, COUNT, %calc

Q9

Customer Segmentation

CASE WHEN on prev_purchases

Q10

Top 3 per Category

CTE + RANK() window

Q11

Repeat Buyer Subscription

CASE WHEN + PARTITION BY

Q12

Revenue by Age Group

CASE WHEN age bins

KEY SQL FINDINGS

Highest-revenue category

Electronics at \$486K - 2.5x the second-ranked Accessories (\$194K)

Gender Split

Male 39.3% | Female 35.4% | Other 25.3% of total revenue

Customer Loyalty

3,085 Loyal customers vs. 1,354 Returning and 561 New

Age Group

51+ segment is top revenue contributor at \$399K, followed by 35-50 at \$266K

Discounts

Discount users show higher avg. purchase (\$219) vs. non-discount (\$182) - attract high-intent buyers

Subscriptions

Subscribed customers avg. spend \$270.58 vs. \$165.72 for non-subscribers

Discount Dependency

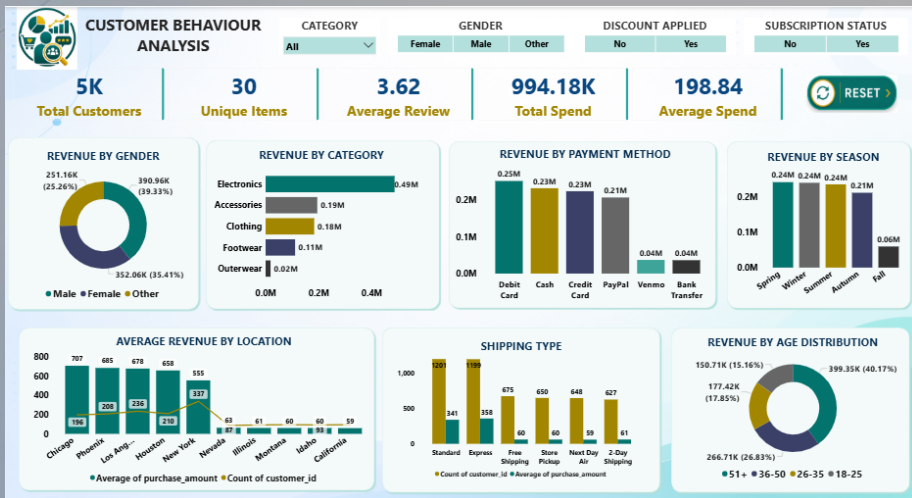
Laptop, Phone, Headphones have >50% of sales with discount applied

Shipping

Express avg. spend \$60.48 vs. Standard \$58.46 - premium shipping = higher spenders

POWER BI DASHBOARD

Page 1 - Customer Overview



Revenue by Season

Spring, Winter & Summer each
~\$240K; Fall lags at \$60K

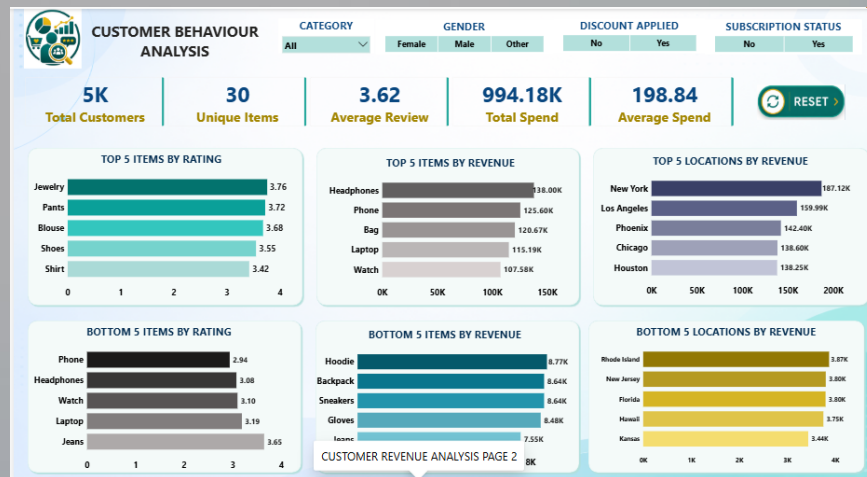
Payment Methods

Debit Card leads (\$250K), followed
by Cash, Credit Card & PayPal
(~\$210-230K each)

Top Locations

Chicago (\$707 avg.), Houston
(\$678), Phoenix (\$485) lead by
average revenue

Page 2 - Rankings & Performance



Top 5 Items by Revenue

- Headphones: \$138K | Phone: \$125K | Bag: \$121K
- Laptop: \$115K | Watch: \$108K

Top 5 Locations by Revenue

- New York (\$187K), Los Angeles (\$160K), Phoenix (\$142K)

Bottom 5 Items by Revenue

- Hoodie (\$8.8K), Backpack (\$8.6K), Sneakers (\$8.6K), Gloves (\$8.5K), Jeans (\$7.6K)

Bottom 5 Locations by Revenue

- Rhode Island (\$3.9K), New Jersey (\$3.8K), Florida (\$3.8K), Hawaii (\$3.8K), Kansas (\$3.5K)

BUSINESS RECOMMENDATIONS

Subscription Program

Promote exclusive benefits (early access, free shipping, loyalty points) to grow subscriber base and CLV

Customer Retention

Introduce loyalty tier system with incentives to move 1,354 Returning customers to Loyal status

Shipping Upsell

Promote Express and 2-Day options - these customers already spend more on average

Payment Adoption

Incentivize PayPal/Venmo to reduce payment friction and increase conversion rates

Discount Strategy

Retain discounts for Electronics (high dependency) but reduce for Clothing/Footwear to protect margins

Product Positioning

Promote top-rated items (Gloves, Sandals, Boots) in campaigns; improve Phone and Headphone ratings

Age-Group Marketing

Focus on 51+ and 35-50 segments (top contributors); use digital channels for 18-35 growth

Seasonal Planning

Align inventory with Spring/Summer peaks; target campaigns to boost Fall revenue (\$60K vs \$240K+ other seasons)

CONCLUSION

This project delivers an end-to-end analytics workflow across 5,000 customer records using Python, MySQL, and Power BI.

Category- Electronics

Drives ~49% of total revenue (\$486K out of \$994K)

Subscriptions

Subscribed customers spend 63% more on average (\$270 vs \$165)

Top Segment

51+ age group is the highest-value segment at \$399K revenue

Discounts

Attract high-intent buyers but create dependency risk in Electronics